

Quartz resonator — Solution

X26 (2026) — English translation

A0

?? pts

Write down the code for the package containing the electronic components on your answer sheets. Check that it is complete.

To complete this step, you need to carefully look around the entire workplace. If you look carefully enough, there is a high probability of finding a package with electronic components on the table, on which an individual code in the “*OOO 00 00*” format is applied. After discovering the package, we carefully rewrite the entire code using a pen into a specially designated place on the answer sheet (this place is easily identified by the inscription “A0” on the left). At this stage you need to be extremely focused, otherwise it may lead to mistakes. The most common mistakes are: rewriting only the last 4 characters (“*00 00*”); rewriting characters in the wrong order (“*00 00 OOO*”); errors in spelling of characters (“*OOO oo oo*”); adding extra characters (“*00 00 (OOO)*”); no code on the answer sheet. The next step in completing this item is to check the completeness of the package. The check procedure is as follows: Open the bag. Carefully pour out all the ingredients from the bag. Make sure the bag is really empty. Familiarize yourself with the list of equipment from the condition (if you are sufficiently insinuating when reading the condition of the problem, it can be found by the inscription Equipment). Check for the presence of a quartz resonator in the poured elements (looks like a gray box with legs). Read the table of capacitor markings presented after the photo of the equipment in the condition. Attention! It is required to compare in order the presence of capacitors in the table (see point 3) and on the table, poured out of the bag in point 1 (see point 1). If the mapping between the names in the table and the capacitors on the table is one-to-one, then you can proceed to the next step. Otherwise, you should contact the duty officer.

To complete this step, you need to carefully look around the entire workplace. If you look carefully enough, there is a high probability of finding a package with electronic components on the table, on which an individual code in the “*OOO 00 00*” format is applied.

After discovering the package, we carefully rewrite the entire code using a pen into a specially designated place on the answer sheet (this place is easily identified by the inscription “A0” on the left). At this stage you need to be extremely focused, otherwise it may lead to mistakes. The most common mistakes are:

The next step in completing this item is to check the completeness of the package. The verification procedure is as follows:

Answer: \textbf{Answer:}

OOO 00 00

A1

0.20 pts

Find the signal frequency $f_{\max}$ at which the voltage amplitude across the quartz resonator is maximum. The difference between $f_{\max}$ and 2 MHz is no more than 1 kHz.

Answer: \textbf{Answer:}

$f_{\max} = 1.999860 \text{ MHz}.$

This means that the impedance module can be recalculated using the following formula:

$$|Z| = \frac{U_{CH2} R_0}{\sqrt{(U_{CH1} - U_{CH2} \cos \varphi)^2 + (U_{CH2} \sin \varphi)^2}}$$

Let's recalculate (see solution A3) and plot the dependence.

$$f_0 = 1.999000 \text{ MHz}$$

$$\tilde{U}_{CH1} = \tilde{I}(R_0 + Z);$$

$$\tilde{U}_{CH2} = \tilde{I}Z.$$

Let's divide the equations and get:

$$\frac{U_{CH2}}{U_{CH1}} e^{i\varphi} = \frac{Z}{R_0 + Z}.$$

Now we can get an expression for the impedance of a quartz resonator:

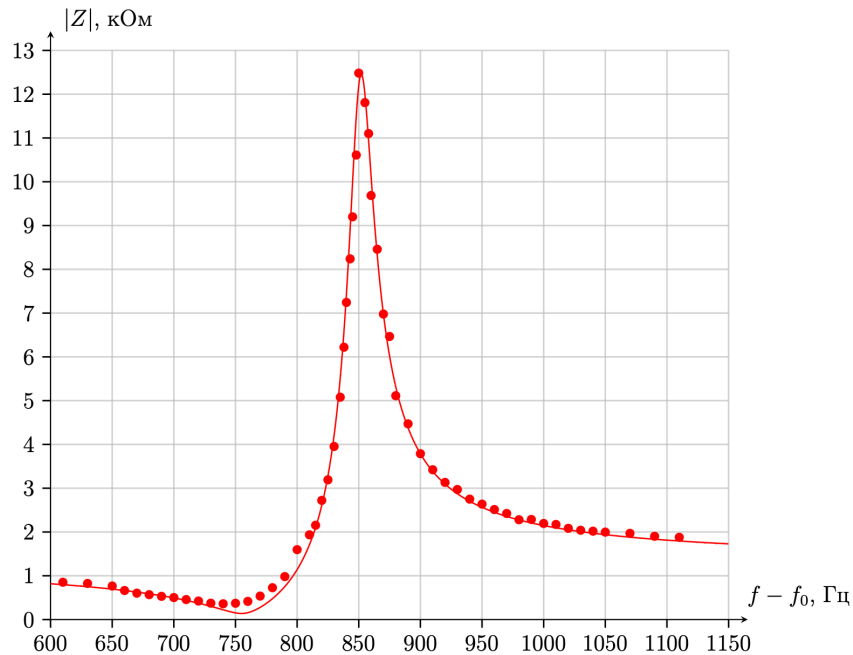
$$Z = U_{CH2} R_0 \cdot \frac{e^{i\varphi}}{U_{CH1} - U_{CH2} e^{i\varphi}}$$

This means that the impedance module can be recalculated using the following formula:

$$|Z| = \frac{U_{CH2} R_0}{\sqrt{(U_{CH1} - U_{CH2} \cos \varphi)^2 + (U_{CH2} \sin \varphi)^2}}$$

Let's perform the recalculation (see solution A3) and plot the dependence.

$$f_0 = 1.999000 \text{ MHz}$$



A5**0.10 pts**

From the plotted graph $|Z|(f)$, find the frequencies f and f at which $|Z|$ is maximum and minimum, respectively.

Answer: \textbf{Answer:}

$$f = 1.999850 \text{ MHz}; \quad f = 1.999740 \text{ MHz}.$$

A6**0.30 pts**

Express $\omega_s, \omega_p, \alpha$ in terms of R, L, C, C_0 .

Impedance of parallel connection of capacitor C_0 and RLC -chain:

$$\frac{1}{Z} = \frac{1}{R + i\omega L + \frac{1}{i\omega C}} + i\omega C_0$$

Let's simplify the expression:

$$\frac{1}{Z} = \frac{1 + i\omega C_0 \left(R + i\omega L + \frac{1}{i\omega C} \right)}{R + i\omega L + \frac{1}{i\omega C}}$$

$$Z(\omega) = \frac{\frac{1}{LC} - \omega^2 + i\omega \frac{R}{L}}{i\omega C_0 \left(\frac{1}{LC_0} + \frac{1}{LC} - \omega^2 + i\omega \frac{R}{L} \right)}$$

Answer: \textbf{Answer:} Thus:

$$\omega_s = \sqrt{\frac{1}{LC}}; \quad \omega_p = \sqrt{\frac{1}{LC} \left(1 + \frac{C}{C_0} \right)}; \quad \alpha = \frac{R}{L}.$$

A7**0.10 pts**

Using the approximations described above, express $|Z|^2(\omega)$ in terms of $\alpha, \omega, \omega_s, \omega_p, C_0$.

Let us take advantage of the fact that for any two complex numbers z_1, z_2 $\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$ holds:

Answer: \textbf{Answer:}

$$|Z|^2 = \frac{(\omega_s^2 - \omega^2)^2 + (\omega_p \alpha)^2}{(\omega_p C_0)^2 ((\omega_p^2 - \omega^2)^2 + (\omega_p \alpha)^2)}.$$

A8**0.20 pts**

Using the approximations described above, express f and f in terms of ω_s, ω_p .

$$\frac{d|Z|^2}{d\omega^2} = \frac{2}{(\omega_p C_0)^2} \cdot \frac{-(\omega_s^2 - \omega^2)((\omega_p^2 - \omega^2)^2 + (\omega_p \alpha)^2) + (\omega_p^2 - \omega^2)((\omega_s^2 - \omega^2)^2 + (\omega_p \alpha)^2)}{((\omega_p^2 - \omega^2)^2 + (\omega_p \alpha)^2)^2}$$

$$\frac{d|Z|^2}{d\omega^2} = \frac{2}{(\omega_p C_0)^2} \frac{(\omega_p^2 - \omega_s^2)((\omega_p \alpha)^2 - (\omega_p^2 - \omega^2)(\omega_s^2 - \omega^2))}{((\omega_p^2 - \omega^2)^2 + (\omega_p \alpha)^2)^2}$$

$$\frac{d|Z|}{d\omega} = \frac{2|Z|}{\omega(\omega_p C_0)^2} \frac{(\omega_p^2 - \omega_s^2)((\omega_p \alpha)^2 - (\omega_p^2 - \omega^2)(\omega_s^2 - \omega^2))}{((\omega_p^2 - \omega^2)^2 + (\omega_p \alpha)^2)^2}$$

Let us find the frequencies at which the derivative $|Z|$ is reset to zero:

$$(\omega_p \alpha)^2 - (\omega_p^2 - \omega^2)(\omega_s^2 - \omega^2) = 0$$

$$-\omega^4 + \omega^2(\omega_p^2 + \omega_s^2) + (\omega_p\alpha)^2 - (\omega_p\omega_s)^2 = 0.$$

Solution of the quadratic equation:

$$\omega = \sqrt{\frac{(\omega_s^2 + \omega_p^2) \pm \sqrt{(\omega_s^2 + \omega_p^2)^2 + 4\omega_p^2(\alpha^2 - \omega_s^2)}}{2}}$$

Using the approximation $\alpha \ll \omega_s$, we obtain the frequencies at which the derivative $|Z|$ is reset: $\omega = \omega_s$ and $\omega = \omega_p$.

Note. In paragraph A12 we will show the validity of this neglect for a given quartz resonator.

Now let's determine which of the two points has resonance and which has antiresonance:

$$|Z|^2(\omega_p) = \frac{1}{(\omega_p C_0)^2} \cdot \frac{(\omega_s^2 - \omega_p^2)^2 + (\omega_p\alpha)^2}{(\omega_p\alpha)^2}$$

$$|Z|^2(\omega_s) = \frac{1}{(\omega_p C_0)^2} \cdot \frac{(\omega_p\alpha)^2}{(\omega_s^2 - \omega_p^2)^2 + (\omega_p\alpha)^2}$$

Now we can see that at $\omega = \omega_s$ there will be antiresonance, and at $\omega = \omega_p$ there will be resonance.

Answer: \textbf{Answer:}

$$f = \frac{\omega_p}{2\pi}; \quad f = \frac{\omega_s}{2\pi}$$

A9

0.45 pts

Remove the dependence of the resonance frequencies f and antiresonance f of the quartz resonator on the external capacitance C_{ext} . As an external capacitance C_{ext} , it is enough to use each of the provided capacitors separately, without combining them.

Answer: \textbf{Answer:} C_{ext} , pF f , MHz f , MHz 0 1.999850 1.999750 6.0 1.999842 1.999755 15 1.999832 1.999755 10 1.999837 1.999755 27 1.999822 1.999755 33 1.999816 1.999755 47 1.999807 1.999755 56 1.999804 1.999755 68 1.999798 1.999755 82 1.999794 1.999755

A10

0.80 pts

Construct a linearized graph of C_{ext} versus f and f .

Let us write the expression for the resonant frequency of a quartz resonator with a parallel connection C_{ext} :

$$\omega_p(C_{ext}) = \sqrt{\frac{1}{LC} \left(1 + \frac{C}{C_0 + C_{ext}} \right)} = \omega_s \sqrt{1 + \frac{C}{C_0 + C_{ext}}}$$

Then after the transformation we can obtain linearization:

$$C_{ext} = \frac{f^2 C}{f^2 - f_s^2} - C_0$$